Lecture 4: Inference for Subpopulation Proportions and Differences

August 27, 2023

Review

- In previous lectures, we discussed learning about population proportions and means based on a random sample (with replacement) from the population.

Review

- In previous lectures, we discussed learning about population proportions and means based on a random sample (with replacement) from the population.
- We saw that for large enough sample sizes, our sample proportion/mean was likely to be close to the population proportion/mean.

Review

- In previous lectures, we discussed learning about population proportions and means based on a random sample (with replacement) from the population.
- We saw that for large enough sample sizes, our sample proportion/mean was likely to be close to the population proportion/mean.
- We also demonstrated that we could predict how close we might by using confidence intervals.

Review

- In previous lectures, we discussed learning about population proportions and means based on a random sample (with replacement) from the population.
- We saw that for large enough sample sizes, our sample proportion/mean was likely to be close to the population proportion/mean.
- We also demonstrated that we could predict how close we might by using confidence intervals.
- We explored a bootstrapping approach and a plug-in approach to producing confidence intervals.

In this lecture

- In this lecture, we discuss learning subpopulation proportions and differences between subpopulation proportions based on a random sample (with replacement) from the population.

In this lecture

- In this lecture, we discuss learning subpopulation proportions and differences between subpopulation proportions based on a random sample (with replacement) from the population.
- Returning to the turnout example, we will consider Black turnout and the difference between Black and non-Black turnout in the 2020 Georgia presidential election.

In this lecture

- In this lecture, we discuss learning subpopulation proportions and differences between subpopulation proportions based on a random sample (with replacement) from the population.
- Returning to the turnout example, we will consider Black turnout and the difference between Black and non-Black turnout in the 2020 Georgia presidential election.
- Even when the overall sample size is fixed (625 for our example), different random samples will produce different subgroup sample sizes, and this adds a layer of complexity.

In this lecture

- In this lecture, we discuss learning subpopulation proportions and differences between subpopulation proportions based on a random sample (with replacement) from the population.
- Returning to the turnout example, we will consider Black turnout and the difference between Black and non-Black turnout in the 2020 Georgia presidential election.
- Even when the overall sample size is fixed (625 for our example), different random samples will produce different subgroup sample sizes, and this adds a layer of complexity.
- This complexity will not complicate our analysis much but it does complicate the justification for the analysis.

2020 Presidential Election Turnout in Georgia

We return to the turnout application, but we are now interested in Black turnout and Black turnout in comparison to non-Black turnout. Recall that ...

- based on a survey of 625 registered voters a week before the election, we got a turnout rate of .68,
- after the election, we found that the turnout rate among all registered voters was .70

2020 Presidential Election Turnout in Georgia

We return to the turnout application, but we are now interested in Black turnout and Black turnout in comparison to non-Black turnout. Recall that ...

- based on a survey of 625 registered voters a week before the election, we got a turnout rate of .68,
- after the election, we found that the turnout rate among all registered voters was .70

Suppose we also want to predict Black turnout and the difference between Black and non-Black turnout.

2020 Presidential Election Turnout in Georgia

We return to the turnout application, but we are now interested in Black turnout and Black turnout in comparison to non-Black turnout. Recall that ...

- based on a survey of 625 registered voters a week before the election, we got a turnout rate of .68,
- after the election, we found that the turnout rate among all registered voters was .70

Suppose we also want to predict Black turnout and the difference between Black and non-Black turnout.

- in our survey of 625 registered voters, 181 were Black with a turnout rate of .73 and 444 were non-Black with a turnout rate of .66.

2020 Presidential Election Turnout in Georgia

We return to the turnout application, but we are now interested in Black turnout and Black turnout in comparison to non-Black turnout. Recall that ...

- based on a survey of 625 registered voters a week before the election, we got a turnout rate of .68,
- after the election, we found that the turnout rate among all registered voters was .70

Suppose we also want to predict Black turnout and the difference between Black and non-Black turnout.

- in our survey of 625 registered voters, 181 were Black with a turnout rate of .73 and 444 were non-Black with a turnout rate of .66.
- after the election, we found that the turnout rate among the 30% of registered voters who were black was .74 and among the 70% of non-Black registered voters was .68.

Four Questions

1. .73 is close to the Black turnout rate of .74. Did we get lucky?

Four Questions

1. .73 is close to the Black turnout rate of .74. Did we get lucky?
2. Could we have predicted how close we would get to .74 before the election?

Four Questions

1. .73 is close to the Black turnout rate of .74. Did we get lucky?
2. Could we have predicted how close we would get to .74 before the election?
3. The survey turnout gap of .07 is close to the population turnout gap of .06. Did we get lucky?

Four Questions

1. .73 is close to the Black turnout rate of .74. Did we get lucky?
2. Could we have predicted how close we would get to .74 before the election?
3. The survey turnout gap of .07 is close to the population turnout gap of .06. Did we get lucky?
4. Could we have predicted how close we would get to a .06 turnout gap before the election?

Sampling Distribution for Black Turnout Proportion

Let the variable X take the value 1 if Black and 0 if non-Black

i	1	2	...	625		
	Y_1, X_1	Y_2, X_2	...	Y_{625}, X_{625}	$\overline{Y}_{625}$	$\frac{\sum_{i=1}^{625} Y_i X_i}{\sum_{i=1}^{625} X_i}$
	1,0	1,1	...	0,0	.68	.69

- Note that $\sum_{i=1}^{625} Y_i X_i$ is not a binomial RV because different samples have different numbers of Black and non-Black individuals.
- To simulate the sampling distribution (or bootstrapped sampling distribution), we repeatedly sample with replacement individuals and their (Y_i, X_i) variable pairs from the population (or the sample).

Sampling Distribution for Black Turnout Proportion

Let the variable X take the value 1 if Black and 0 if non-Black

i	1	2	...	625		
s	Y_1, X_1	Y_2, X_2	...	Y_{625}, X_{625}	$\overline{Y}_{625}$	$\frac{\sum_{i=1}^{625} Y_i X_i}{\sum_{i=1}^{625} X_i}$
1	1,0	1,1	...	0,0	.68	.69
2	0,0	1,0	...	1,1	.70	.72

- Note that $\sum_{i=1}^{625} Y_i X_i$ is not a binomial RV because different samples have different numbers of Black and non-Black individuals.
- To simulate the sampling distribution (or bootstrapped sampling distribution), we repeatedly sample with replacement individuals and their (Y_i, X_i) variable pairs from the population (or the sample).

Sampling Distribution for Black Turnout Proportion

Let the variable X take the value 1 if Black and 0 if non-Black

i	1	2	...	625		
s	Y_1, X_1	Y_2, X_2	...	Y_{625}, X_{625}	$\overline{Y}_{625}$	$\frac{\sum_{i=1}^{625} Y_i X_i}{\sum_{i=1}^{625} X_i}$
1	1,0	1,1	...	0,0	.68	.69
2	0,0	1,0	...	1,1	.70	.72
3	1,0	1,1	...	0,1	.75	.76
4	1,1	1,0	...	0,0	.73	.73
5	0,0	0,0	...	1,1	.69	.71
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
1 mil	1,0	0,1	...	1,1	.74	.75
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

- Note that $\sum_{i=1}^{625} Y_i X_i$ is not a binomial RV because different samples have different numbers of Black and non-Black individuals.
- To simulate the sampling distribution (or bootstrapped sampling distribution), we repeatedly sample with replacement individuals and their (Y_i, X_i) variable pairs from the population (or the sample).

Simulating the Sampling Distribution

ID	1	2	3	4	...	...	...	...	7,233,584	Mean
y_{ID}	0	1	0	1	...	...	...	...	1	.70
x_{ID}	0	0	1	0	...	...	...	...	1	.30

For each row of the sampling distribution,

```
dfprop = data.frame(y,x)
ind = sample(1:7233584, size=625, replace=TRUE)
dfind = df[ind,]
subpopMean = mean(dfind$y[dfind$x == 1])
# alternative approach
#subpopMean = sum(dfind$y*dfid$x)/sum(dfind$x)
```

Simulating the Bootstrapped Sampling Distribution

i	1	2	3	4	...	625	$\bar{y}_{625}$
y_i	1	1	0	1	...	0	.68

Starting with the sample (first row of sampling distribtution), each row of the bootstrapped sampling distribution is generated as the following:

```
bootind = sample(1:625, size=625, replace=TRUE)
dfboot = dfind[bootind,]
subpopMean = mean(dfboot$y[dfboot$x == 1])
# alternative approach
#subpopMean = sum(dfboot$y*dfboot$x)/sum(dfboot$x)
```